Rathi Ghosh

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Professional Summary

Results-driven Full-Stack & Mobile Developer specializing in React Native (Expo), Next.js, and Supabase. Proven ability to architect and deliver production-ready platforms from scratch, prioritizing secure data flow, automated workflows, and scalable database design. Equipped with a strong foundation in Data Structures, Algorithms, and core computer science principles, dedicated to building high-performance applications.

Education

Narula Institute of Technology July 2023 – July 2027
B.Tech in Computer Science and Engineering, CGPA: 8.21 North 24 Parganas, West Bengal

Uttarpara Model School (H.S.) – Class XII 2022 – 2023
West Bengal Council of Higher Secondary Education, 66.7%

Uttarpara Model School – Class X 2020 – 2021
West Bengal Board of Secondary Education, 83.84%

Projects

Xpert – Online Examination and Learning Platform GitHub
A production-grade, full-stack educational ecosystem for independent teachers.

- **Architected** cross-platform web and mobile portals using Next.js and Expo, implementing secure role-based access control (RBAC) via Supabase Auth for distinct teacher and student environments
- **Engineered** a frictionless join-code batch enrollment system, eliminating manual roster tracking and streamlining onboarding for multiple concurrent classrooms
- **Developed** a dynamic examination engine featuring a media-rich question builder (MCQ, True/False, Numerical, Theoretical, Image) and a persistent, reusable question bank
- **Designed** a robust server-side evaluation algorithm supporting granular positive/negative marking rules, completely automating the grading pipeline and eliminating manual teacher evaluation
- **Fortified** exam integrity by implementing strict server-side validation and state management to prevent duplicate submissions while allowing safe session recovery for active attempts
- **Automated** the generation of instant PDF result reports featuring detailed, question-level analytics for comprehensive student performance tracking
- **Enforced** strict data privacy across 7 relational PostgreSQL tables utilizing Supabase Row Level Security (RLS), guaranteeing isolated data access and preventing unauthorized data leaks

Technical Skills

Languages: TypeScript, SQL, Java
Frontend & Mobile: React Native (Expo), Next.js, React, Tailwind CSS
Backend: Node.js, REST APIs, PostgresSQL
Auth & Security: JWT, Role-Based Access Control (RBAC)
Tools & Workflow: Git, GitHub